

SOLUTIONS

1) Determine the derivative of each function.

a) $y = \sin(4x)$

$$y' = \cos(4x)(4)$$

$$y' = 4\cos(4x)$$

b) $f(x) = \sin(2x + \pi)$

$$f'(x) = \cos(2x + \pi)(2)$$

$$f'(x) = 2\cos(2x + \pi)$$

c) $y = -2\sin(3\theta)$

$$y' = -2\cos(3\theta)(3)$$

$$y' = -6\cos(3\theta)$$

d) $y = \sin^2 x$

$$y = (\sin x)^2$$

$$y' = 2(\sin x)(\cos x)$$

$$y' = 2 \sin x \cos x$$

e) $f(x) = \cos^2 x - \sin^2 x$

$$f(x) = (\cos x)^2 - (\sin x)^2$$

$$f'(x) = 2\cos x(-\sin x) - 2\sin x(\cos x)$$

$$f'(x) = -2\cos x \sin x - 2\sin x \cos x$$

$$f'(x) = -2(\cos x \sin x + \sin x \cos x)$$

$$f'(x) = -2(2\sin x \cos x)$$

$$f'(x) = -2\sin(2x)$$

f) $y = 3\sin^2(2t - 4) - 2\cos^2(3t + 1)$

$$y = 3[\sin(2t - 4)]^2 - 2[\cos(3t + 1)]^2$$

$$y' = 3(2)[\sin(2t - 4)]\cos(2t - 4)(2) - 2(2)[\cos(3t + 1)][-\sin(3t + 1)](3)$$

$$y' = 12\sin(2t - 4)\cos(2t - 4) + 12\cos(3t + 1)\sin(3t + 1)$$

$$y' = 6\sin[2(2t - 4)] + 6\sin[2(3t + 1)]$$

$$y' = 6\sin(4t - 8) + 6\sin(6t + 2)$$

$$g) f(t) = \sin^2(\cos t)$$

$$P(t) = [\sin(\cos t)]^2$$

$$f'(t) = 2[\sin(\cos t)][\cos(\cos t)](-\sin t)$$

$$f'(t) = -2\sin(t)\sin(\cos t)\cos(\cos t)$$

$$h) f(x) = -x^2 \sin(3x - \pi)$$

$$f'(x) = -2x \sin(3x - \pi) + [\cos(3x - \pi)](3)(-x^2)$$

$$f'(x) = -2x \sin(3x - \pi) - 3x^2 \cos(3x - \pi)$$

$$i) f(\theta) = \sin^2 \theta \cos^2 \theta$$

$$f(\theta) = (\sin \theta)^2 (\cos \theta)^2$$

$$f'(\theta) = 2\sin \theta \cos \theta \cos^2 \theta + 2\cos \theta (-\sin \theta) \sin^2 \theta$$

$$f'(\theta) = 2\sin \theta \cos^3 \theta - 2\cos \theta \sin^3 \theta$$

$$j) y = x^{-1} \cos^2 x$$

$$y = x^{-1} (\cos x)^2$$

$$y' = -1x^{-2} (\cos^2 x) + 2\cos x (-\sin x) x^{-1}$$

$$y' = \frac{-\cos^2 x}{x^2} - \frac{2\cos x \sin x}{x}$$

$$k) y = 2 \tan x - \tan(2x)$$

$$y' = 2\sec^2 x - \sec^2(2x)(2)$$

$$y' = 2\sec^2 x - 2\sec^2(2x)$$

$$l) y = (\tan x + \cos x)^2$$

$$y' = 2(\tan x + \cos x)(\sec^2 x - \sin x)$$

2) Find the slope of the function $y = 2 \cos x \sin(2x)$ at $x = \frac{\pi}{2}$.

$$y' = -2 \sin x [\sin(2x)] + \cos(2x)(2)(2 \cos x)$$

$$y' = -2 \sin x [\sin(2x)] + 4 \cos(2x) \cos x$$

$$y'(\frac{\pi}{2}) = -2 \sin(\frac{\pi}{2}) [\sin(\pi)] + 4 \cos(\pi) \cos(\frac{\pi}{2})$$

$$y'(\frac{\pi}{2}) = -2(1)(0) + 4(-1)(0)$$

$$y'(\frac{\pi}{2}) = 0$$

$$m = 0$$

3) Find the equation of the line that is tangent to $y = x^2 \sin(2x)$ at $x = -\pi$.

Slope: $y' = 2x \sin(2x) + \cos(2x)(2)x^2$

$$y' = 2x \sin(2x) + 2x^2 \cos(2x)$$

$$y'(-\pi) = 2(-\pi) \sin(-2\pi) + 2(-\pi)^2 \cos(-2\pi)$$

$$y'(-\pi) = -2\pi(0) + 2\pi^2(1)$$

$$y'(-\pi) = 2\pi^2$$

Point:

$$y(-\pi) = (-\pi)^2 \sin(-2\pi)$$

$$y(-\pi) = 0$$

$$(-\pi, 0)$$

Eqn: $y = mx + b$

$$0 = 2\pi^2(-\pi) + b$$

$$b = 2\pi^3$$

$$y = 2\pi^2 x + 2\pi^3$$

4) Determine $\frac{d^2y}{dx^2}$ for $y = x^2 \cos x$.

$$\frac{dy}{dx} = 2x \cos x + (-\sin x)(x^2)$$

$$\frac{dy}{dx} = 2x \cos x - x^2 \sin x$$

$$\frac{d^2y}{dx^2} = 2 \cos x + (-\sin x)(2x) - [2x \sin x + \cos x (x^2)]$$

$$= 2 \cos x - 2x \sin x - 2x \sin x - x^2 \cos x$$

$$= 2 \cos x - 4x \sin x - x^2 \cos x$$

5)a) Write $y = \csc x$ in terms of $\sin x$ as a reciprocal function.

$$y = \frac{1}{\sin x}$$

b) Write the function in terms of a negative power of $\sin x$

$$y = (\sin x)^{-1}$$

c) Use the power rule and chain rule to find the derivative of $y = \csc x$

$$y' = -1(\sin x)^{-2}(\cos x)$$

$$y' = \frac{-\cos x}{\sin^2 x}$$

$$y' = \frac{-\cos x}{(\sin x)(\sin x)}$$

$$y' = -\cot x \left(\frac{1}{\sin x}\right)$$

$$y' = -\cot x \csc x$$

Answers:

1)a) $\frac{dy}{dx} = 4 \cos(4x)$ b) $f'(x) = 2 \cos(2x + \pi)$ c) $\frac{dy}{d\theta} = -6 \cos(3\theta)$ d) $\frac{dy}{dx} = 2 \sin x \cos x$

e) $f'(x) = -2 \sin(2x)$ f) $\frac{dy}{dx} = 12 \sin(2t - 4) \cos(2t - 4) + 12 \cos(3t - 1) \sin(3t - 1)$

g) $f'(t) = -2 \sin(\cos t) \cos(\cos t) \sin t$ h) $f'(x) = -3x^2 \cos(3x - \pi) - 2x \sin(3x - \pi)$

i) $f'(\theta) = -2 \sin^3 \theta \cos^3 \theta + 2 \sin \theta \cos^3 \theta$ j) $\frac{dy}{dx} = \frac{-2}{x} \cos x \sin x - \frac{\cos^2 x}{x^2}$

k) $y' = 2 \sec^2 x - 2 \sec(2x)$ l) $y' = 2(\tan x + \cos x)(\sec^2 x - \sin x)$

2) 0

3) $y = 2\pi^2 x + 2\pi^3$

4) $\frac{d^2 y}{dx^2} = -x^2 \cos x - 4x \sin x + 2 \cos x$

5)a) $y = \frac{1}{\sin x}$ b) $y = (\sin x)^{-1}$ c) $\frac{dy}{dx} = -\csc x \cot x$